Harry J. Greatorex

h.greatorex@qub.ac.uk | 0000-0002-5302-0887 | HGreatorex

Astrophysics Research Centre - Queen's University Belfast, Belfast, United Kingdom, BT7 1NN

Employment

- **Postdoctoral Research Fellow - Queen's University Belfast** *April 2025 - Present*

- Formulated and executed independent research on the physics of solar flares, with a primary focus on EUV Late-Phase (ELP) emission, flare energetics, and multi-wavelength diagnostics spanning EUV, SXR, HXR, and radio wavelengths.

- Led the largest statistical study of EUV Late-Phase flares to date (2010–2025), based on automated and manual analysis of SDO/AIA, SDO/EVE, and GOES observations, enabling detailed examination of ELP occurrence, timing, and energetics [*Manuscript accepted May 2026*].

- Developed Python pipelines for large-scale flare analysis, including automated data ingestion, background correction, peak detection, feature extraction, and event classification, designed for reproducibility and application to multi-thousand-event samples.

- Conducted spectroscopic diagnostics of coronal plasma properties including velocity, broadening, electron density, and compositional analysis of post-flare active regions associated with ELP emission. Combined spatio-temporal analyses using multi-wavelength imaging data to detect evidence of secondary reconnection in EUV late-phase events via inflow signatures, thermal variations, and loop system evolutions.

- Conceptualised, advertised, and supervised an undergraduate research intern project in solar flare analysis.

- Collaborated with geophysics and space weather experts as solar physics specialist in consolidated analyses of the Sun-Earth relationship focussed on the impact of flare-related emission on the Earth's ionosphere. [*Items 2 & 3 in Publications*]

Education

- **Queen's University Belfast - PhD Solar Physics** *October 2021 - May 2025*

  Thesis: *Observational Analysis of Lyman-Alpha Emission in Solar Flares*
  Supervisors: *Dr. Ryan Milligan and Prof. Mihalis Mathioudakis*

- **Queen's University Belfast - MSci Physics (Integrated - 4 Years)** *September 2017 - June 2021*

  Project: *Computational Modelling of the Chromospheric Response to Solar Flare Heating in RADYN* **[1st Class Hons]**
  Supervisors: *Prof. Mihalis Mathioudakis and Dr. Peter Keys*

Publications

[1] **Greatorex H.J**, O'Hare A.N., Bekker S., Campbell R.C, Keane D.C., and Milligan R.O. *Solar Physics.* (Accepted)
The EUV Late-Phase: Statistical Results from 15 Years of Solar Dynamics Observatory Observations.

[2] Bekker S., O'Hare A.N., **Greatorex H.J** *JGR Space Physics.* (In Prep)
A Statistical Study of the Ionospheric Total Electron Content Response to the Late Phase of a Solar Flare.

[3] O'Hare A.N., Bekker S.,**Greatorex H.J**, and Milligan R.O. *Atmosphere.* (2025)
Investigating a Characteristic Time Lag in the Ionospheric F-Region's Response to Solar Flares.

[4] Majury L., Milligan R.O, Butler E., **Greatorex H.J**, and Kazachenko M. *Solar Physics.* (2025)
Spectral Irradiance Variability in Lyman-Alpha Emission During Solar Flares.

[5] **Greatorex H.J**, Milligan R.O., Dammasch I.E. *Solar Physics.* (2024)
On the Instrumental Discrepancies in Lyman-alpha Observations of Solar Flares.

[6] **Greatorex H.J**, Milligan R.O., Chamberlin P.C. *Astrophysical Journal.* (2023)
Observational Analysis of Lyman-alpha Emission in Equivalent Magnitude Solar Flares.

Funding

- **STFC Long Term Attachment Funding** *January 2023 []*

Conducted four months of overseas fieldwork at NASA Goddard Space Flight Centre (Maryland, USA) under the supervision of Dr. Brian Dennis, Dr. Graham Kerr, and Dr. Joel Allred. This visitation allowed me to significantly enhance my expertise in HXR Spectral Analysis using OSPEX, leading to publication of (Greatorex et al. 2023)

KEY CONFERENCES & MEETINGS

- **SDO 2025 Science Workshop - Boulder, Colorado** *February 2025*
  Gave a talk on the instrumental effects on EUV measurements across different observing regimes, calibration techniques, and vantage points from missions such as SDO, ASO-S, Solar Orbiter, MAVEN, GOES, and PROBA2 - Next Horizon: the Future Solar and Heliophysics Missions. (Greatorex et al. 2024).

- **Hinode/IRIS/SPHERE Joint Science Meeting - MSU Bozeman, Montana** *July 2024*
  Gave a talk on the relationship between nonthermal electrons and Lyman-alpha emission in Solar Flares. (Greatorex et al. 2023). Also attended secondary meeting amongst the active Max Millennium Chief Observers to discuss the future of the program.

- **SoSpIM Science Meeting - PMOD/ETH Davos, Swizterland** *June 2024*
  Invited to give a talk on active and past projects to members of the SoSpIM instrument team. Contributed to discussion of future science goals of SoSpIM. Planned collaborative projects using SoSpIM and adjacent Lyman-alpha instruments.

- **Solar Orbiter/Parker Solar Probe/DKIST Joint Meeting - SWRI San Antonio, Texas** *April 2024*
  Presented a poster on the relationship between nonthermal electrons and Lyman-alpha emission in Solar Flares. (Greatorex et al. 2023).

- **EVE Science Meeting - CU Lasp Boulder, Colorado [Online]** *July 2023*
  Gave an online presentation to members of the SDO/EVE team on the relationship between nonthermal electrons and Lyman-alpha emission in Solar Flares. (Greatorex et al. 2023).

SEMINARS

- **Queen's University Belfast ARC Seminar** *November 2023*
  Observational Analysis of Lyman-alpha Emission in Equivalent Magnitude Solar Flares [30 min]

- **NASA Goddard Heliophysics Division Seminar [*Invited*]** *March 2023*
  Observational Analysis of Lyman-alpha Emission in Equivalent Magnitude Solar Flares [30 min]

ADDITIONAL SCIENTIFIC EXPERIENCE

- **ISSI Team Member** *January 2025 - Present*

  Member of an ISSI team aiming to examine the Sun-Earth connection through the relationship between solar flares and other transient phenomena and the Earth's ionosphere. Leader of a research project to investigate the physical mechanisms that drive the ELP using spectrally and spatially resolved observations.

- **Session Chair - Hinode/IRIS Joint Science Meeting - UCL, UK** *June 2025*

  Invited to chair the session "Energy and Mass Transfer throughout the Solar Atmosphere" and assist the LOC in the organisation of the joint science meeting.

- **Max Millennium Chief Observer** *October 2023 - Present*

  Member of the Max Millennium Observing Programme. Regularly carry out flare forecasting using data from space-based and ground-based observatories to predict solar activity in the upcoming 24 hour period. The Max Millennium messages reach a wide global audience including mission teams, research institutions, and independent scientists.

- **SoSpIM Science Team Member** *June 2024 - Present*

  Member of the team dedicated to the scientific development of the SoSpIM instrument commisioned for the Solar-C mission headed by JAXA.

- **FlaMEX Science Team Member** *May 2025 - Present*

  Member of the team formulated for the development of an ESA Mini-F mission proposal contributing directly to the proposal and consulting on scientific concepts within the project conceptualisation. *Proposal Outcome: Shortlisted*.

TEACHING & MENTORSHIP EXPERIENCE

- **Supervisor - QUB Summer Research Internship** *July 2025 - September 2025*
  Designed and supervised an intern project, "*Chasing the Afterglow: EUV Late-Phase Studies with SDO*", examining the role of imaging and spectroscopic measurement discrepancies in the search for the EUV Late-Phase following solar flares. This project ran for 8 weeks and allowed the student to develop computational analysis skills and successfully contribute to a scientific publication.

- **MSci Student Mentor** *January 2022 - Present*
  Supervised several MSci research projects within the solar flare group. All students went on to achieve a 2nd class or higher with some undertaking a PhD in the field.

- **Pathway Opportunity Programme Demonstrator** *June 2023*
  Ran computer programming class for sixth form students from disadvantaged backgrounds. Developed a Python workbook to guide students through analysis of SOHO/LASCO and SDO/AIA data to calculate the velocity of a CME.

Outreach Experience

- **Invited Speaker - U3A** *March 2026*
  Invited to give a scientific outreach talk to an astronomy and astrophysics group affiliated with the U3A organisation.
  Talk Title: Don't Look Back Into The Sun - A Journey Through the Observational Discovery and Understanding of Solar Flares
  Slides available on request.
- **NI Science Festival** *February 2022 - Present*
  Assisting in the organisation and fulfillment of the annual Northern Ireland Science Festival. This includes running stands for the astronomy day and making content for videos.
- **Girls in Maths & Physics** *June 2024*
  Ran a successful activity stand for high school students to "predict a solar flare" as part of the outreach event.

Computational Skills

Programming Languages: Python (incl. sunpy, scikit-learn, stixpy), IDL (incl. SolarSoft), C, LaTeX, SQL
Programming Skills: Object Oriented Programming, HXR Spectroscopy, Image Processing, Instrument Cross-calibration, Large Data Handling, High-Performance Computing, Machine Learning.

Software Projects

- **Package: pydesat-aia.py** *April 2026* []
  Developed desaturation tools for SDO/AIA images translated from SSWIDL to Python based on the methods of Schwartz. (2015) and Kazachenko et al. (2017).

Current Research Projects

- **Searching for QPP in ELP Emission as an Inidcator of Conitnued Reconnection**
  Examining spectrally resolved EUV data to identify the presence of quasi-periodic pulsations in their timeseries and comparing the periods to known ranges that correspond to periodic reconnection, thus identifying the potential driver of the secondary EUV emission.
- **Resolving the Multithermal Structure and Dynamics of the EUV Late-Phase**
  Performing spectroscopic analysis of EUV Late-Phase flare loops using Hinode/EIS to diagnose plasma properties and dynamics. This includes measurements of Doppler velocities, non-thermal line broadening, and calculation of electron density to characterise the thermal and dynamic evolution of ELP structures. Carried out diagnostics of cooling timescales and emission measures. Comparisons with main phase flare loops are used to assess whether Late-Phase loops are formed and heated under distinct physical conditions, including prolonged heating, wave activity, or sustained mass transport.
- **Human Trained Agent for MMCO Flare Forecasting**
  Developing machine learning methods to improve upon the daily flare forecasts currently provided by Max Millennium Cheif Observers. This project utilities large volumes of archival data typically used for flare forecasting by the human counterpart, evaluates the historical success of the human observer across 15-years of forecasts, and uses this to mimic and subsequently improve upon the forecasts. These improved forecasting capabilities will be combined into a consolidated signle agent that can produce live 'Human-Like' forecasts of solar activity. This will be distributed as a tool to the wider community.

References

1. **Dr. Ryan Milligan**
   Senior Lecturer
   Astrophysics Research Centre - Queen's University Belfast
   Email: r.milligan@qub.ac.uk
   Line Manager and former PhD Supervisor
2. **Prof. Dr. Louise Harra**
   Director - Physikalisch-Meteorologisches Observatorium Davos/World Radiation Center &
   Affiliated Professor - ETH Zurich
   Email: Louise.Harra@pmodwrc.ch
   Collaborator